

UID EXP: 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

- 1) Setting Up HTML File
- 2) Create the JavaScript File for Charts

1) SETTING UP HTML FILE

```
< index.html X JS script.js
index.html > html > head > style > canvas
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
6   <title>Inventory Management Visualization</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      text-align: center;
11      margin: 50px;
12    }
13    canvas {
14      margin: 20px auto;
15    }
16  </style>
17 </head>
18 <body>
19   <h1>Inventory Management System</h1>
20   <canvas id="pieChart" width="400" height="400"></canvas>
21   <canvas id="barChart" width="400" height="400"></canvas>
22
23   <!-- Chart.js CDN -->
24   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
25   <!-- External JavaScript file -->
26   <script src="script.js"></script>
27 </body>
28 </html>
29
```

2) Create the JavaScript File for Charts

```
index.html JS scriptjs X
JS scriptjs > ...
1 // script.js
2
3 // Inventory data
4 const inventoryData = {
5   labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
6   datasets: [{
7     label: 'Items in Stock',
8     data: [200, 150, 100, 80, 50],
9     backgroundColor: [
10      '#FF6384',
11      '#36A2EB',
12      '#FFCE56',
13      '#4BC0C0',
14      '#9966FF'
15    ],
16   }]
17 };
18
19 // Pie Chart Configuration
20 const ctxPie = document.getElementById('pieChart').getContext('2d');
21 const pieChart = new Chart(ctxPie, {
22   type: 'pie',
23   data: inventoryData,
24   options: {
25     responsive: true,
26     plugins: {
27       title: {
28         display: true,
29         text: 'Inventory Distribution'
30       }
31     }
32   }
33 });
34
```

```
34
35 // Bar Chart Configuration
36 const ctxBar = document.getElementById('barChart').getContext('2d');
37 const barChart = new Chart(ctxBar, {
38   type: 'bar',
39   data: inventoryData,
40   options: {
41     responsive: true,
42     plugins: {
43       title: {
44         display: true,
45         text: 'Items in Stock by Category'
46       }
47     },
48     scales: {
49       y: {
50         beginAtZero: true
51       }
52     }
53   }
54 });
55
```

3) OUTPUT THE VALIDATION:

Inventory Management System

